

3.4 HVAC ROUGH-IN GUIDE

Project Information

Property Address: _____

Permit Number: _____ Inspection Date: _____

HVAC Contractor: _____ License #: _____

Phone: _____

SYSTEM OVERVIEW

System Component	Type/Specification	Notes
Heating System Type	☐ Forced Air ☐ Radiant ☐ Heat Pump	
Heating Source	☐ Gas ☐ Electric ☐ Oil ☐ Heat Pump	
Cooling System Type	☐ Central AC ☐ Heat Pump ☐ Mini-Split	
System Capacity - Heating	_____ BTU	
System Capacity - Cooling	_____ Tons	
Furnace/Air Handler Location	☐ Basement ☐ Attic ☐ Closet ☐ Garage	
Condensing Unit Location	Location: _____	
Zoned System	☐ Yes - _____ zones ☐ No	
Thermostat Type	☐ Programmable ☐ Smart ☐ Manual	
Number of Thermostats	_____ thermostats	

LOAD CALCULATION

Manual J Load Calculation Performed: ☐ Yes ☐ No

Calculated By: _____ Date: _____

Total Heating Load: _____ BTU

Total Cooling Load: _____ BTU (or _____ Tons)

Load Calculation Documents Attached: ☐ Yes ☐ No

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DUCTWORK LAYOUT - SUPPLY

✓	Location/Room	Register Size	CFM Required	Duct Size	Notes
☐	Main trunk line	N/A			
☐	Master Bedroom				
☐	Master Bathroom				
☐	Bedroom #2				
☐	Bedroom #3				
☐	Bedroom #4				
☐	Bathroom #2				
☐	Living Room				
☐	Dining Room				
☐	Kitchen				
☐	Family Room				
☐	Office/Den				
☐	Hallway(s)				
☐	Basement/Lower Level				
☐	Bonus Room				
☐	Other: _____				

Supply Ductwork Notes:



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DUCTWORK LAYOUT - RETURN AIR

✓	Location	Return Grille Size	Duct Size	Notes
☐	Main return trunk	N/A		
☐	Central hallway return			
☐	Master bedroom return			
☐	Bedroom #2 return			
☐	Living area return			
☐	Basement return			
☐	Second floor return			
☐	Other: _____			

Return Air Sizing Notes:

Total Return Air CFM Required: _____ CFM

Return air adequately sized for system: ☐ Yes ☐ NoReturn air located away from supply registers: ☐ Yes ☐ No**EQUIPMENT LOCATIONS**

✓	Equipment/Item	Location	Notes
☐	Furnace/Air Handler		
☐	Equipment clearances verified	Per manufacturer specs	
☐	Access for service/filter change	Adequate access provided	
☐	Condensing unit pad location		
☐	Condensing unit clearances	Min. 24" one side, 12" others	
☐	Thermostat location - Zone 1		
☐	Thermostat location - Zone 2		
☐	Thermostat on interior wall	Not on exterior wall	
☐	Thermostat away from heat sources	Not near windows, vents, etc.	

☐	Zone dampers (if applicable)	Locations: _____	
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UTILITIES & CONNECTIONS REQUIRED

Gas Line (if applicable)

✓	Item	Specification	Notes
☐	Gas line size to furnace	Per BTU load/distance	
☐	Gas shut-off valve at equipment	Within 6' of unit	
☐	Gas line properly supported	Per code	
☐	Sediment trap installed	Drip leg at appliance	
☐	Gas line pressure tested	☐ Yes ☐ Pending	

Electrical Requirements

✓	Item	Specification	Notes
☐	Furnace/Air Handler circuit	_____ amp dedicated	
☐	Condensing unit circuit	_____ amp dedicated	
☐	Disconnect switch at condensing unit	Within sight of unit	
☐	Thermostat wiring - 18/5 or greater	Common wire included	
☐	Zone control wiring (if applicable)	Per system specs	

Condensate Drain

✓	Item	Specification	Notes
☐	Primary condensate drain line	3/4" PVC typical	
☐	Drain line properly sloped	1/4" per foot minimum	
☐	Drain terminates to approved location	Floor drain, exterior, etc.	
☐	P-trap in condensate line	Required for proper operation	
☐	Secondary/emergency drain pan (if req.)	For attic install	
☐	Emergency drain visible termination	Alert to primary drain clog	

Combustion Air & Venting (Gas/Oil Systems)

✓	Item	Specification	Notes
☐	Combustion air provisions adequate	Per manufacturer/code	
☐	Vent pipe size correct	Per appliance specs	
☐	Vent termination location approved	Clearances from openings	
☐	Vent properly supported/sloped	Per manufacturer	

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ROUGH-IN QUALITY CHECKLIST

Ductwork Installation

✓	Item	Notes
☐	All ductwork installed per approved plan	
☐	Duct sizes match design specifications	
☐	Ductwork properly supported every 4'-6'	
☐	All joints sealed with mastic or approved tape	
☐	No cloth duct tape used (not code compliant)	
☐	Flexible duct fully extended (no sagging)	
☐	Flexible duct runs limited to 8'-10' max	
☐	Proper transition fittings used	
☐	Return air plenums sealed at equipment	
☐	No return air leaks in unconditioned spaces	
☐	Supply boots properly installed and sealed	
☐	Boot supports installed in ceiling/floor	
☐	Register/grille locations marked	
☐	Ductwork in unconditioned spaces insulated	
☐	Insulation R-value meets code minimum	
☐	Vapor barrier on insulation facing outward	
☐	No sharp bends or kinks in ductwork	
☐	Turning vanes in square elbows (if required)	

Ductwork Notes:



ROUGH-IN QUALITY CHECKLIST (CONTINUED)

Equipment Rough-In

✓	Item	Notes
☐	Equipment platform/stand level and secure	
☐	Vibration isolation if required	
☐	Refrigerant line set path planned/installed	
☐	Line set properly sized for system	
☐	Line set insulated and protected	
☐	Line set penetrations sealed	
☐	Thermostat wire run to all locations	
☐	Extra thermostat wire coiled at equipment	
☐	Condensate drain rough-in complete	
☐	Equipment access door/panel clearance verified	

Penetrations & Sealing

✓	Item	Notes
☐	All duct penetrations through fire-rated walls sealed	
☐	Duct penetrations through roof properly flashed	
☐	Duct penetrations to exterior sealed airtight	
☐	No ducts running through garage (if code prohibits)	
☐	Fire dampers installed where required	
☐	Backdraft dampers on exhaust vents	

Code Compliance

✓	Item	Notes
☐	System meets energy code requirements	
☐	Equipment SEER/AFUE ratings meet minimums	
☐	Duct insulation meets energy code	
☐	Combustion air meets code requirements	
☐	Equipment clearances meet manufacturer specs	© 2024 Build Your House
☐	Gas venting meets code and manufacturer specs	

☐	Return air not from garage or bathrooms	
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ADDITIONAL HVAC COMPONENTS

Ventilation & Exhaust

✓	Item/Location	Specification	Notes
☐	Whole-house ventilation system	☐ ERV ☐ HRV ☐ Exhaust ☐ N/A	
☐	Kitchen range hood exhaust duct	To exterior, _____ CFM	
☐	Bathroom exhaust - Master	To exterior, _____ CFM	
☐	Bathroom exhaust - Bath #2	To exterior, _____ CFM	
☐	Bathroom exhaust - Bath #3	To exterior, _____ CFM	
☐	Laundry room exhaust (if req.)	To exterior	
☐	Dryer vent duct	4" rigid to exterior	
☐	Dryer vent length within limits	Max length per code	
☐	All exhaust ducts terminate outside	Not into attic/crawlspace	
☐	Exhaust vent terminations have caps	Pest/weather protection	

Air Quality & Filtration

✓	Item	Notes
☐	Filter location/access planned	
☐	Filter size: _____ x _____ x _____	
☐	Upgraded filtration system (if applicable)	☐ HEPA ☐ Electronic ☐ Media ☐ Standard
☐	UV light system (if applicable)	Location: _____
☐	Humidifier rough-in (if applicable)	☐ Whole-house ☐ Bypass ☐ Steam
☐	Humidifier water line to equipment	1/4" line with shutoff
☐	Dehumidifier provisions (if applicable)	

General Notes: _____



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INSPECTION RESULTS & CORRECTIONS**Initial HVAC Rough-In Inspection**

Inspection Date: _____ Time: _____

Inspector Name: _____

Inspector Badge/ID: _____

Result: ☐ PASSED ☐ FAILED - Corrections Required**Deficiencies/Code Violations Found:****Corrections Made**

Date Corrections Completed: _____

HVAC Contractor: _____ License #: _____

Description of Corrections:**Re-Inspection (if required)**

Re-Inspection Date: _____ Time: _____

Inspector Name: _____

Result: ☐ PASSED ☐ FAILED - Additional Corrections Required**Additional Comments:**

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Final Sign-Off

Owner/Builder Signature: _____ Date: _____

HVAC Contractor Signature: _____ Date: _____

Inspector Signature: _____ Date: _____

Permit Number: _____ Final Approval: ☐ YES